AgriHGT: Heterogeneous Graph Transformer for Cross-Crop Leaf Disease Relation Modeling in Tomato, Papaya, Zucchini, and Bottle Gourd

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***Abstract*—Leaf-disease classification across multiple crops is challenging because visually similar symptoms can occur across different disease classes, while relationships among crops, diseases, and broader disease groups are not explicitly represented by conventional image-only classifiers. This study presents AgriHGT, a two-stage visual-relational framework that combines pretrained visual representations with a heterogeneous graph containing leaf-image, crop, disease, and disease-category nodes. The framework was evaluated on the Agri-Vision4 dataset, covering Tomato, Papaya, Zucchini, and Bottle Gourd across 28 crop-specific classes. After duplicate removal, 26,866 images were retained and partitioned using a duplicate-aware and support-controlled splitting procedure. Seven supervised and self-supervised visual backbones were evaluated. Extracted image embeddings were used as leaf-image node features and processed through relation-aware Heterogeneous Graph Transformer (HGT) message passing, followed by residual visual fusion and disease prediction. The highest observed performance was obtained with DINOv3 ViT-B/16 and two HGT layers, achieving 93.87% test accuracy and 85.88% macro-F1. The corresponding feature-only logistic-regression reference achieved 91.63% accuracy and 80.27% macro-F1. In the main DINOv3 depth experiment, performance increased progressively from zero to two HGT layers, providing evidence that message passing was useful for this visual representation. Component ablation produced mixed effects, with the residual visual pathway, learned fusion gate, and disease visual grounding showing useful contributions, while some structural relations did not consistently improve performance. The HGT experiments were conducted in a transductive setting, with validation and test nodes present in the graph during training but without access to their disease labels. These findings show that combining self-supervised visual representations with heterogeneous graph modelling is promising for multi-crop leaf-disease classification, while further inductive and multi-seed evaluation is needed to establish broader generalization.**

***Index Terms*—Plant disease classification, heterogeneous graph transformer, HGT, DINOv3, self-supervised learning, graph neural network, multi-crop classification.**

I. Introduction

Plant disease classification from leaf images is an important computer-vision problem because plant diseases can reduce crop yield, lower crop quality, and cause substantial economic loss to farmers. Early and accurate disease identification is therefore important for timely treatment and effective crop management. However, manual diagnosis often depends on farmers, agricultural officers, or plant pathology experts and can become difficult when expert support is unavailable, when different diseases exhibit visually similar symptoms, or when large numbers of plants must be examined. Automated leaf-disease classification can provide useful decision support by identifying disease conditions directly from leaf images. Deep-learning-based plant-disease diagnosis has therefore received considerable attention as a means of supporting faster and more scalable disease identification [1].

Plant diseases commonly produce visible changes in leaf colour, texture, shape, spots, lesions, holes, and other damaged regions. These visual characteristics have made leaf images suitable for deep-learning-based disease recognition. Convolutional Neural Networks (CNNs) have been widely used for this task because they can automatically learn discriminative visual features instead of relying on manually designed descriptors. CNN-based methods have achieved strong performance on several plant-disease datasets [2], [3]. Transfer learning has also been widely adopted, where models pretrained on large image datasets are adapted to plant-disease recognition. Architectures such as ResNet, DenseNet, EfficientNet, and MobileNet have been successfully applied in this setting [4], [5]. Recent work has also continued to improve plant-disease recognition through combinations of convolutional feature extraction and attention-based modelling [6].

More recently, Vision Transformers (ViTs) have been investigated for plant-disease recognition. Unlike conventional CNNs, a ViT divides an image into patches and applies self-attention to model relationships among them, allowing information from spatially distant image regions to be considered jointly. ViT-based and hybrid CNN–transformer models have demonstrated strong performance on plant-disease datasets [7]–[10]. Hierarchical architectures such as Swin Transformer have also been explored to capture both local and global visual information in plant-disease recognition [11]. In addition, self-supervised learning has received increasing attention because it can learn strong visual representations while reducing dependence on manually labelled data [12], [13]. Related studies have also examined cross-dataset and unseen-crop recognition, showing that maintaining disease-discriminative representations across changing crop and image domains remains difficult [14], [15].

Another important research direction is multi-crop disease classification. Instead of developing a separate model for each crop species, a single classifier can include multiple crops and their associated diseases. Recent studies have developed multi-crop systems covering crops such as rice, wheat, bean, tea, banana, sugarcane, tomato, potato, and others [16]–[18]. Such systems are useful because one model can support a broader range of crops and disease classes. Cross-crop recognition has also been studied using semantic disease attributes and unseen-crop domain generalization, indicating that information shared across crop species may help recognition when disease concepts overlap [15], [19]. However, most existing multi-crop approaches still formulate the task as conventional image classification. Each image is processed independently by a CNN or transformer and assigned to one class, while relationships among the crop, its possible diseases, visually similar leaves, and broader disease groups are generally not represented explicitly.

This limitation is important because plant diseases are not independent concepts. Each disease occurs on a particular host crop, while different diseases may belong to broader biological or symptom-related groups such as fungal, bacterial, viral, pest-related, or nutrient-deficiency conditions. Similar disease symptoms may also occur across different crops. Fine-grained plant-disease studies have shown that separating classes with similar lesion, colour, and texture patterns remains challenging when visual differences are subtle [20]–[22]. Consequently, useful information may exist not only within an individual leaf image but also in the relationships among images, crops, diseases, and disease categories. Conventional image classifiers do not explicitly represent this relational structure.

Graph Neural Networks (GNNs) provide a natural mechanism for modelling such relationships. In a graph, entities are represented as nodes and their relationships as edges, allowing connected nodes to exchange information through message passing. Graph-based methods have already been applied to plant-disease problems, including regional feature aggregation, CNN–GCN integration, multimodal visual–graph fusion, and leaf-to-leaf relational modelling [23]–[26].Agricultural knowledge graphs have also been developed to organize relationships among crops, diseases, pests, symptoms, and other agricultural concepts [27], [28]. Knowledge graphs have further been combined with neural and language-based reasoning for agricultural disease analysis [29]. Other studies have combined visual representations with knowledge-graph embeddings for disease classification [30].

Despite this progress, existing approaches generally address only part of the problem. Image-based GNN methods commonly construct graphs among image regions or visually similar leaf images, whereas agricultural knowledge graphs primarily support knowledge storage, retrieval, question answering, or semantic reasoning. Knowledge-graph fusion approaches may concatenate graph embeddings with visual features, but individual leaf images, crops, diseases, and disease categories are not typically represented together as distinct node types within a unified relation-aware message-passing graph. Thus, multi-crop visual classification and structured agricultural knowledge are still often treated as separate components. Recent graph-based and heterogeneous attention approaches further demonstrate the growing interest in combining visual representations with structured relational reasoning for plant-disease diagnosis [31]–[33]. However, these developments do not remove the need to examine a graph in which individual observed leaf images and agricultural semantic entities are represented jointly as different node types.

A further challenge is determining whether any improvement produced by a graph-based model actually comes from relational message passing. A complex graph architecture may outperform a simpler classifier because of additional parameters, projections, or fusion components rather than because of the graph itself. Controlled comparisons are therefore necessary between feature-only classification, architectures without message passing, and models with different graph depths. Component-level ablations are also needed to determine the effects of image similarity, crop information, disease relations, disease taxonomy, residual visual information, and learned feature fusion.

To address these limitations, this study proposes *AgriHGT*, a two-stage visual–relational framework for multi-crop leaf-disease classification. In the first stage, pretrained visual backbones are adapted to the leaf-disease domain using only the training partition and are subsequently used to extract fixed visual embeddings. Seven backbones are evaluated: ResNet50, DenseNet121, EfficientNet-B3, ConvNeXt-Tiny, Swin-Tiny, DINOv2 ViT-B/14, and DINOv3 ViT-B/16. These models represent conventional CNNs, modern CNN architectures, supervised vision transformers, and self-supervised vision transformers.

In the second stage, the extracted image embeddings are incorporated into a heterogeneous graph containing four node types: `leaf_image`, `crop`, `disease`, and `disease_category`. Typed relations represent crop membership, crop-disease associations, disease-category membership, image similarity, selected image-disease connections, and relationships among diseases belonging to the same broader category. A Heterogeneous Graph Transformer (HGT) [34] is then applied so that information associated with different node and relation types can be processed separately during message passing. The HGT representation is combined with

a residual visual pathway through a learnable feature-wise fusion mechanism before final disease prediction.

The main objectives of this study are:

1) To develop a heterogeneous multi-crop disease-classification framework that jointly represents leaf images, crops, diseases, and disease categories.
2) To evaluate seven supervised and self-supervised visual backbones as representations for heterogeneous graph learning.
3) To determine the contribution of HGT message passing by comparing zero, one, and two layer configurations with a feature-only reference model.
4) To investigate the contribution of individual graph and architectural components through controlled ablation experiments.
5) To examine how relational information behaves across multiple crop species within a shared heterogeneous graph.

The main contributions of this study are:

1) A heterogeneous graph framework, named *AgriHGT*, is developed to jointly represent leaf images, crops, diseases, and disease categories using multiple node and relation types within a unified graph.
2) The proposed framework integrates adapted visual embeddings, relation-aware HGT message passing, a residual visual pathway, and a learnable feature-wise fusion mechanism for multi-crop leaf-disease classification.
3) Seven pretrained visual backbones are systematically evaluated within the same experimental framework, including ResNet50, DenseNet121, EfficientNet-B3, ConvNeXt-Tiny, Swin-Tiny, DINOv2 ViT-B/14, and DINOv3 ViT-B/16. Which allowed supervised and self-supervised visual representations to be compared under consistent conditions.
4) Controlled HGT-depth experiments are conducted using zero, one, and two layer configurations to examine the contribution of relational message passing beyond the underlying visual representation.
5) Component-level ablation experiments are performed to investigate the effects of image-similarity relations, crop information, disease relations, disease-category information, residual visual features, meta-node identity, and the learnable fusion mechanism.
6) The study evaluates, in a transductive setting, heterogeneous relational modelling across Tomato, Papaya, Zucchini, and Bottle Gourd within a shared multi-crop graph, providing evidence on how visual and semantic relationships behave across multiple crop species.

The remainder of this paper is organised as follows. Section II reviews CNN-based, transformer-based, self-supervised, multi-crop, GNN, and knowledge-graph approaches to plant-disease recognition. The methodology section describes the dataset preparation, duplicate-aware splitting procedure, backbone adaptation, feature extraction, heterogeneous graph construction, AgriHGT architecture, training procedure, reference models, and ablation settings. The results and discussion section presents backbone comparisons, HGT-depth experiments, class-level performance, qualitative analysis, and component-ablation results. Finally, the conclusion summarises the main findings, limitations, and directions for future work.

II. Related Work

A. CNN and Supervised Transfer-Learning Approaches

Shahoveisi et al. [3] examined transfer learning for rust detection under field and greenhouse conditions. Their dataset contained 1,764 images of sunflower, dry bean, and field-pea leaves, including 907 healthy and 857 rust-infected samples collected under varying backgrounds and lighting conditions. Xception, ResNet50, EfficientNetB4, and MobileNetV2 were tested with different optimizers and learning rates. The best configuration was further evaluated using ten-fold cross-validation. EfficientNetB4 obtained an average accuracy of 94.29%, while the selected configuration achieved $94.34\% \pm 0.010$ under cross-validation. Grad-CAM analysis showed that the model focused on rust pustules rather than unrelated background regions. Although the study included several crop hosts and realistic imaging conditions, the task was limited to binary rust detection and did not model relationships among crops and disease concepts.

Shafik et al. [4] developed a deep transfer-learning ensemble for 15 selected PlantVillage disease classes. Nine pretrained networks, including DenseNet201, ResNet101, ResNet50, EfficientNetB7, NASNetMobile, and ConvNeXtSmall, were used as feature extractors. The extracted features were combined through logistic-regression-based early fusion and lead-voting ensembles. The proposed PDDNet-AE and PDDNet-LVE models achieved accuracies of 96.74% and 97.79%, respectively. The ensemble improved classification performance across the selected classes, although the experiments were based mainly on the controlled PlantVillage setting and required several pretrained networks.

Ali et al. [5] fine-tuned EfficientNet-B0 for apple leaf-disease classification and also evaluated the model on PlantVillage. Their training procedure included stratified splitting, class weighting, full-network fine-tuning, dropout, regularization, and geometric and photometric augmentation. Test accuracies of 99.69% on the apple subset and 99.78% on PlantVillage were reported. The model also retained relatively low memory and FLOP requirements. The study demonstrated that a compact pretrained CNN can provide strong performance for plant-disease recognition, although crop and disease relations were not represented separately from the image classes.

Mahapatra et al. [14] studied cross-dataset generalization using an attention-based CNN for corn and potato diseases. Experiments were conducted on Digipathos, Northern Leaf Blight, PlantVillage, PlantDoc, and CD&S datasets. Both intra-dataset and cross-dataset settings were considered, and background-removed CD&S images were used to examine the influence of background variation. The highest intra-dataset accuracy was 99.38% for potato images from PlantVillage, whereas the strongest average cross-dataset accuracy for corn

was 82.93%. The difference between the two settings illustrates the difficulty of maintaining classification performance when image acquisition conditions change.

B. Vision Transformer and CNN-Transformer Approaches

Borhani et al. [7] compared CNN, Vision Transformer, and combined CNN–ViT architectures using Wheat Rust Classification, Rice Leaf Disease, and PlantVillage datasets. Models were trained from scratch at several image resolutions to examine classification performance, computational complexity, and prediction speed. The ViT configurations generally achieved the strongest classification results, although the attention-based models required greater prediction time. Their transformer modelling remained limited to relationships among image patches within a single image rather than relationships among separate crop or disease entities.

Thakur et al. [8] proposed a lightweight CNN–transformer architecture and evaluated it on five public plant-disease datasets. The model combined early VGG16 convolutional blocks, an Inception module, and four transformer encoder modules while retaining approximately 0.85 million trainable parameters. It achieved 98.86% accuracy on PlantVillage and 89.24% on the more difficult Embrapa dataset. Grad-CAM and LIME were used to examine image regions associated with the predictions. The work showed that convolutional and transformer features can be combined efficiently, but crop–disease classes were still learned as individual classification targets without an explicit relational structure.

Hemalatha and Jayachandran [35] introduced the Plant Disease Localization and Classification Vision Transformer (PDLC-ViT) for joint disease localization and classification. PlantVillage was used for classification, whereas PlantDoc and corresponding masks were used for localization. Their framework incorporated affine augmentation, hyperparameter tuning, early stopping, co-scale representations, co-attention, and cross-attention. Reported results included 99.97% classification accuracy, 99.18% mean average precision, and 99.11% mean average recall. Grad-CAM-based activation maps were also presented. The use of different datasets for localization and classification, however, introduced differences between the two evaluation domains.

Aboelenin et al. [36] combined CNN feature extraction with Vision Transformer attention for apple and corn disease classification. Features from VGG16, Inception-v3, and DenseNet201 were combined with ViT-based self-attention and an MLP classifier. Images were resized to 128×128 and augmented during training. The framework achieved 99.24% accuracy for apple diseases and 98.00% for corn diseases. The approach combined local CNN features with global transformer attention, although it relied on several visual backbones and treated the two crop datasets independently without an explicit relational model connecting crop and disease entities.

C. Self-Supervised, Zero-Shot, and Cross-Domain Recognition

Wang et al. [12] investigated self-supervised representation learning using a masked autoencoder. Their experiments used the CCMT field dataset and an independently collected potato dataset. A masked autoencoder was combined with a ViT backbone, GRU, and CBAM attention module. Rotation, random cropping, and horizontal flipping were applied during training. The model obtained approximately 95.4% accuracy on CCMT and 99.6% on the collected potato dataset. The study showed that self-supervised pretraining can provide useful representations for plant-disease recognition while reducing dependence on labelled examples. The final disease prediction, however, remained image based without an explicit representation of crop–disease relationships.

Zhao et al. [13] proposed a contrastive self-supervised framework with domain adaptation. PlantVillage, PlantDoc, and an augmented leaf-disease dataset were used to construct a setting containing distribution-shifted unlabelled data and a smaller amount of labelled target data. Contrastive pretraining was followed by a domain-adaptation layer and supervised fine-tuning. The highest reported accuracy was 90.52%. Additional experiments examined the effects of dataset size and source–target mixing ratios. Their results showed the value of representation learning under label scarcity and domain shift, although the method did not explicitly model relationships among individual images, crops, and disease concepts.

Kumar et al. [19] examined zero-shot cross-crop disease recognition using a PlantVillage subset containing tomato, potato, and pepper classes. Thirteen manually defined semantic attributes described properties such as colour, shape, texture, spots, and discoloration. These attributes were concatenated with pretrained CNN or ViT features. In a three-class tomato-to-potato experiment, ResNet18 achieved 99.95% accuracy when semantic attributes were included, compared with 62.73% without them. Performance decreased substantially in a more difficult four-class setting, where ViT-B/16 achieved 69.45%. The study demonstrated that semantic disease information can improve cross-crop recognition, although manually defining the attributes limits scalability and the semantic information was combined with visual features rather than propagated through a graph.

Wu et al. [15] studied recognition in unseen crop domains using Text-guided feature Disentanglement Contrast (TDC). PlantVillage was reorganized into six disease concepts shared across eight crop domains. The resulting dataset contained 21,521 original images and 19,369 background-removed training images. Their method used a dual-branch ViT-B/16 encoder, a frozen CLIP text encoder, contrastive disentanglement, domain-adversarial learning, semantic text anchors, and prototype-based classification. Seen-domain accuracy reached 98.04%, while unseen-domain accuracy was 74.29%, with a harmonic mean of 84.53%. Evaluation on an external strawberry powdery-mildew dataset produced 81.99% accuracy. The study directly examined transfer across crop domains, although its formulation depended on disease concepts shared among the selected crops and did not construct an explicit graph linking leaf images, crops, and diseases.

D. Multi-Crop Disease Classification

Subburaj et al. [16] proposed Plantention, a lightweight classifier designed to handle several crops within one model.

TABLE I
TAXONOMY OF REPRESENTATIVE RELATED WORK BY MODEL TYPE, INPUT MODALITY, NODE HETEROGENEITY, MESSAGE PASSING, AND EVALUATION SETTING

Study	Model Type	Input Modality	Heterogeneous Nodes	Graph Message Passing	Evaluation Setting
Shahoveisi et al. [3]	CNN (transfer learning)	Image only	No	No	Inductive
Shafik et al. [4]	CNN ensemble + LR fusion	Image only	No	No	Inductive
Ali et al. [5]	CNN (EfficientNet-B0)	Image only	No	No	Inductive
Mahapatra et al. [14]	Attention-based CNN	Image only	No	No	Inductive (cross-dataset)
Borhani et al. [7]	CNN / ViT / CNN–ViT	Image only	No	No	Inductive
Thakur et al. [8]	CNN–Transformer hybrid	Image only	No	No	Inductive
Hemalatha & Jayachandran [35]	Multitask ViT (PDLC-ViT)	Image only	No	No	Inductive
Aboelenin et al. [36]	CNN–ViT hybrid	Image only	No	No	Inductive
Wang et al. [12]	SSL (MAE + ViT + GRU + CBAM)	Image only	No	No	Inductive
Zhao et al. [13]	Contrastive SSL + domain adapt.	Image only	No	No	Inductive
Kumar et al. [19]	CNN/ViT + semantic attributes	Image + manual attributes	No	No	Inductive
Wu et al. [15]	ViT + frozen CLIP text encoder	Image + text	No	No	Inductive
Subburaj et al. [16]	MobileNetV2 + dual attention	Image only	No	No	Inductive
Hasan et al. [17]	MobileNetV2 (transfer learning)	Image only	No	No	Inductive
Zidan et al. [18]	Custom lightweight CNN	Image only	No	No	Inductive
Bera et al. [23]	GCN (PND-Net)	Image regions	No (single node type)	Yes (GCN)	Inductive
Zhao et al. [29]	LLM + KG + GNN	Text / knowledge graph	Yes	Yes (GNN)	KG reasoning
Yan et al. [27]	NLP + entity extraction (KG)	Text / knowledge graph	Yes	No	KG retrieval
Alwan & Alturfi [30]	CNN/ViT + Node2Vec KG fusion	Image + knowledge graph	Yes	No joint MP	Inductive
Surekha & Subha Mastan Rao [26]	CNN + GNN (AdjLeafGNN)	Leaf-image graph	No (single node type)	Yes (GNN)	Inductive
AgriHGT (this work)	**Visual backbone + HGT**	**Image + relational graph**	**Yes (leaf, crop, disease, disease-category)**	**Yes (relation-aware HGT)**	**Transductive**

Five public datasets covering rice, wheat, tea, banana, and sugarcane were used. The architecture combined a MobileNetV2 encoder with dual split-attention and residual classification modules and contained approximately 7.3 million parameters. The reported accuracy, precision, recall, and F1-score were 98.34%, 98.74%, 99.19%, and 99.02%, respectively. Plantention demonstrated that multiple crop species can be handled by a single lightweight classifier, but the relationships among diseases associated with different crops were not explicitly represented.

Hasan et al. [17] developed a single transfer-learning model for diseases of rice, wheat, and bean. Their combined dataset contained 7,037 annotated leaf images collected from Bangladeshi crop-disease sources and a bean-disease dataset. Images were resized to 224×224, normalized, and augmented through random rotation, horizontal flipping, and zooming. MobileNetV2 was fine-tuned using a 70/15/15 training, validation, and test split and obtained 97.87% test accuracy. The unified classifier reduced the need to maintain separate models for each crop, although the crop–disease combinations were still treated as classification classes rather than connected entities.

Zidan et al. [18] presented a larger multi-crop benchmark containing approximately 45,000 original RGB images collected from several public repositories. The curated dataset contained 51 healthy and diseased classes covering 12 crops. Duplicate and low-quality samples were manually screened, stratified splitting was applied, and augmentation was restricted to the training data. Eight compact pretrained transformers were benchmarked before the development of PlantaNetLite and PlantaNet, which contained approximately 1.28 and 2.58 million parameters, respectively. PlantaNet achieved 99.37% validation accuracy and 99.66% test accuracy. Grad-CAM and Grad-CAM++ were used to examine disease-related regions. The work provided a broad multi-crop classification benchmark, although the individual crop–disease classes were learned without an explicit semantic graph connecting them.

E. Graph Neural Networks and Knowledge-Graph Approaches

Bera et al. [23] introduced PND-Net, which used graph learning to aggregate regional visual information. Banana and coffee nutrition-deficiency datasets, together with potato and PlantDoc disease datasets, were used for evaluation. CNN features were summarized through spatial-pyramid pooling and represented as graph nodes before GCN-based aggregation. Data augmentation included random rotation, scaling, Gaussian blur, and random cropping. The model achieved 96.18% accuracy on the potato dataset and 84.30% on PlantDoc. Grad-CAM and t-SNE were used to inspect the learned visual representations. In this framework, the graph nodes represented image-derived regions rather than distinct agricultural entities such as leaf images, crops, diseases, and disease categories.

Zhao et al. [29] combined large language models, an agricultural knowledge graph, representation learning, GNN components, neural-symbolic reasoning, and graph attention for disease and pest identification in *Elaeagnus angustifolia*. The graph-attention configuration obtained precision of 0.94, recall of 0.92, and accuracy of 0.93. The knowledge structure supported semantic reasoning over agricultural information. The study focused on one plant species and was designed around knowledge-assisted disease and pest identification rather than heterogeneous multi-crop leaf-image classification.

Yan et al. [27] constructed CropDP-KG from text and associated information in a Chinese crop diseases and pests image–text database. Natural-language processing and entity extraction were used to identify eight primary entity types and seven relation types. The resulting graph contained 13,840 entities and 21,961 relationships. CropDP-KG was intended for knowledge querying, question answering, overview, and knowledge management. It therefore provides structured agricultural knowledge, but image embeddings are not treated as graph nodes and the graph is not used as a GNN-based leaf-image classifier.

Alwan and Alturfi [30] combined visual features with structured knowledge using EfficientNet-B8, a Vision Transformer, and knowledge-graph fusion. Their dataset contained approx-

imately 87,000 RGB images across 38 crop–disease classes. Crop types, environmental factors, and disease relationships were included in a knowledge graph, and Node2Vec-style graph representations were concatenated with CNN and transformer visual features. The method reported 99.3% accuracy, 98.9% precision, 98.92% recall, and 98.67% F1-score. Grad-CAM was used to examine image-based predictions. This work is closely related to the use of relational information in plant-disease recognition, but individual observed leaf images were not represented as heterogeneous graph nodes and graph information was fused with visual features rather than propagated jointly through an HGT.

Surekha and Subha Mastan Rao [26] proposed AdjLeafGNN for leaf-to-leaf relational modelling. A 10,000-image subset of PlantVillage covering tomato, potato, grape, and apple was normalized and resized to 224×224. Horizontal and vertical flipping, $\pm 15^{\circ}$ rotation, and brightness variation were used for augmentation. CNN embeddings were treated as nodes, and K-nearest-neighbour edges were constructed with $K = 5$ according to feature similarity. The combined attention-enhanced CNN and GNN achieved $98.88\% \pm 0.33\%$ classification accuracy and $98.71\% \pm 0.32\%$ F1-score. Disease-spread prediction reached an AUC-ROC of 0.942. The method directly modelled relationships among leaf images, although all leaf nodes belonged to the same node type and crop, disease, and disease-category entities were not represented as separate nodes.

F. Comparative Summary of Related Work

The reviewed studies approach plant-disease recognition from several directions. CNN and transformer models learn visual patterns from individual leaf images, while self-supervised methods focus on improving visual representations when labelled data are limited. Multi-crop classifiers extend image classification to several crop species within one model. These approaches generally retain an image-to-class formulation and do not explicitly encode relationships among the crop, disease, and broader disease group associated with each sample.

Graph-based studies introduce relational information in different forms. Some methods construct graphs from image regions or connect leaf images according to visual similarity. Agricultural knowledge graphs represent semantic relations among crops, diseases, pests, symptoms, and other entities, while knowledge-graph fusion methods combine these representations with visual features. The reviewed work therefore provides evidence for both visual and relational modelling, but the two are commonly handled through separate mechanisms.

AgriHGT differs in the structure of the classification graph. Individual leaf images, crops, diseases, and disease categories are represented as distinct node types, while different relations are processed through relation-aware HGT message passing. The model also separates the visual pathway from the graph representation and combines them through learnable feature-wise fusion. Its experimental design compares feature-only classification with zero, one, and two layer HGT configurations and examines individual graph and fusion components through ablation. The resulting setting allows the role of relational information to be studied within a single heterogeneous multi-crop classification graph rather than treating all crop-disease combinations only as independent output classes.

III. Materials and Methods

A. Overview of the Methodology

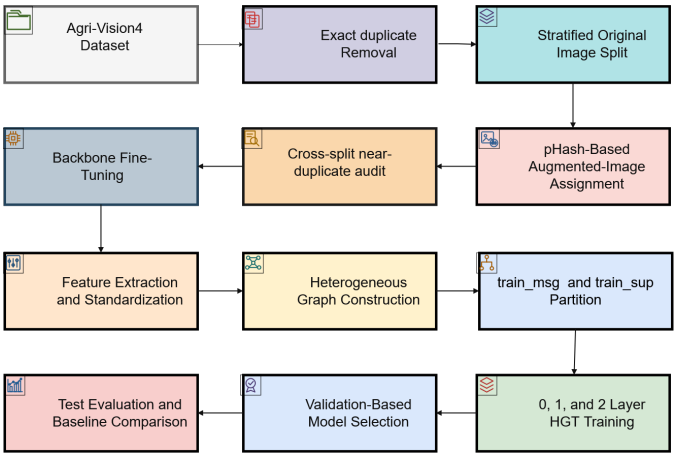


Fig. 1. Overview of the AgriHGT methodology.

AgriHGT was implemented as a two-stage visual-relational classification framework. In the first stage, a pretrained visual backbone was partially fine-tuned on the training partition and then used to extract one fixed feature vector from each retained leaf image. In the second stage, these visual features were assigned to `leaf_image` nodes in a heterogeneous graph containing crop, disease, and disease-category nodes. Relation-aware message passing was performed using the Heterogeneous Graph Transformer (HGT) [34]. The resulting graph representation was combined with a residual visual representation before disease prediction. The complete experimental pipeline is shown in Fig. 1.

B. Dataset

The Agri-Vision4 dataset [37] was used in this study. It contains leaf images from four crops: Tomato, Papaya, Zucchini, and Bottle Gourd, covering 28 crop-specific healthy and disease classes. The images were collected from agricultural fields in Bangladesh under natural lighting conditions and were validated by agricultural experts. The published dataset contains both original images and augmented images produced through transformations including rotation, flipping, brightness and contrast adjustment, and noise addition. The distribution of original and augmented images across the four crops is shown in Fig. 2. A large proportion of the dataset consists of augmented images, which increases the number of training samples available for the different crops.

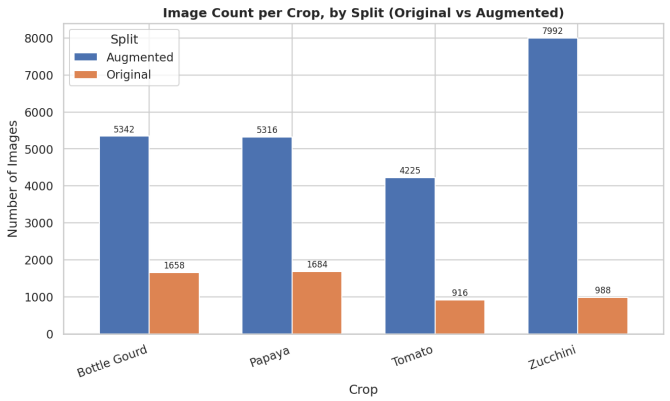


Fig. 2. Distribution of original and augmented images across the four crops in the Agri-Vision dataset.

Before duplicate removal, the dataset contained 28,121 image files. The distribution of the original images across the 28 classes is reported in Table II. The class frequencies were unequal, so classification accuracy was reported together with macro-F1 to provide an evaluation that gives equal importance to minority and majority classes.

C. Exact-Duplicate Removal

Exact duplicate removal was performed before constructing the training, validation, and test partitions. An MD5 hash was calculated for each of the 28,121 image files. Files with identical hashes were grouped, and one representative image from each group was retained.

A total of 1,255 byte-identical duplicate files were removed. The resulting dataset contained 26,866 images, consisting of 4,304 original images and 22,562 augmented images.

D. Duplicate-Aware Initial Split

A random split over the complete set of original and augmented images could place an original image in one partition and a transformed version of the same image in another. Such overlap could produce overly optimistic validation or test results. The data was splitted using the original images first.

Class-stratified sampling with a 70/15/15 training, validation, and test ratio was first applied to the original images. The initial split contained 3,004 training images, 650 validation images, and 650 test images.

Perceptual hashes were then calculated using

$$\texttt{hash_size} = 16,$$

which produces a 256-bit perceptual hash. Each augmented image was compared with original images belonging to the same crop and class. If the minimum Hamming distance was no greater than six, the augmented image was linked to the closest original image and assigned to the same partition.

Using this criterion, 1,552 augmented images were linked to original images. The remaining 21,010 augmented images could not be confidently associated with a particular original image and were initially assigned to the training partition.

E. Post-Hoc Near-Duplicate Audit

After the initial split, a same-class near-duplicate audit was performed across the training, validation, and test partitions. Perceptual hashes were compared for original-original, original-augmented, and augmented-augmented image pairs.

When two same-class images from different partitions had a perceptual-hash Hamming distance of six or less, the validation or test-side image was moved to the training partition. The audit was repeated until no further cross-partition pair satisfying this condition remained. The successive audit rounds identified 41, 11, 3, and finally 0 violating pairs.

F. Per-Class Support Control

Because duplicate removal and the near-duplicate audit could remove the only validation or test sample of a minority class, a support-control step was applied after the audit.

If a class had no validation sample or no test sample, an eligible training image from that class was reassigned to the missing partition. The near-duplicate audit was then repeated. If the reassigned image introduced a new cross-partition near-duplicate relation, it was returned to the training partition and another eligible image was considered.

The final split contained 25,122 training images, 896 validation images, and 848 test images. At least one validation and one test image were retained for each of the 28 classes. The same final split was used for every backbone experiment.

The resulting partition is described as duplicate-aware or leakage-controlled rather than completely leakage-free. The audit was restricted to same-class comparisons and to similarities captured by the selected perceptual-hash representation and Hamming-distance threshold. The supported claim is therefore that no same-class cross-partition pair was detected at a perceptual-hash Hamming distance of six or less.

G. Visual Backbone Selection

The visual backbone is an important part of AgriHGT because the feature extracted from each leaf image is used as the initial representation of the corresponding `leaf_image` node. The same feature space is also used to construct the `similar_to` relations between leaf images. Therefore, the graph can change when a different backbone is used.

Seven visual backbones were selected to cover different convolutional, transformer, supervised, and self-supervised representation-learning approaches used in recent plant-disease classification studies [8], [16], [17].

- ResNet50;
- DenseNet121;
- EfficientNet-B3;
- ConvNeXt-Tiny;
- Swin-Tiny;
- DINOv2 ViT-B/14; and
- DINOv3 ViT-B/16.

The purpose of using several backbone families was not only to identify the backbone with the highest classification performance. The main aim was also to examine whether heterogeneous graph learning behaves similarly when the initial image representation is changed.

TABLE II
DISTRIBUTION OF ORIGINAL IMAGES ACROSS THE 28 CROP-SPECIFIC CLASSES BEFORE DUPLICATE REMOVAL

Crop	Class	Original Images
Bottle Gourd	Alternaria Leaf Blight	303
Bottle Gourd	Anthracnose	276
Bottle Gourd	Downy Mildew	286
Bottle Gourd	Early Alternaria Leaf Blight	179
Bottle Gourd	Fungal Damage Leaf	39
Bottle Gourd	Healthy	260
Bottle Gourd	Mosaic Virus	315
Papaya	Bacterial Blight	183
Papaya	Carica Insect Hole	318
Papaya	Curled Yellow Spot	538
Papaya	Healthy Leaf	189
Papaya	Mosaic Virus	119
Papaya	Pathogen Symptoms	286
Papaya	Yellow Necrotic Spots Holes	51
Tomato	Downy	57
Tomato	Healthy	288
Tomato	Mosaic	195
Tomato	Spot	311
Tomato	White Spot	65
Zucchini	Angular Leaf Spot	120
Zucchini	Anthracnose	129
Zucchini	Downy Midew	153
Zucchini	Dry Leaf	67
Zucchini	Healthy	88
Zucchini	Insect Damage	78
Zucchini	Iron Chlorosis Damage	65
Zucchini	Xanthomonas Leaf Spot	86
Zucchini	Yellow Mosaic Virus	202
	Total	**5,246**

1) ResNet50: ResNet50 was selected as a standard convolutional baseline. Residual connections allow deep convolutional layers to be trained while reducing the difficulty caused by vanishing gradients. ResNet-based transfer learning has also been used for plant-disease recognition [3], [4].

2) DenseNet121: DenseNet121 was selected to represent a different form of convolutional feature learning. Instead of residual addition, DenseNet connects each layer with the outputs of earlier layers. This encourages feature reuse and allows low-level and high-level visual information to remain available throughout the network.This behaviour is useful for leaf-disease images because both small visual details, such as spots and lesions, and larger texture patterns may be important.

3) EfficientNet-B3: This backbone was included because plant-disease symptoms can appear at different spatial scales, while EfficientNet architectures have shown strong performance in leaf-disease classification [5].

4) ConvNeXt-Tiny: ConvNeXt-Tiny was selected as a modern convolutional architecture. Although it remains a CNN, its design includes several ideas influenced by the development of vision transformers. It therefore provides an important middle point between conventional CNNs and transformer-based models. Its inclusion helps determine whether improvements associated with newer visual representations require a transformer architecture or can also be obtained from a modernized convolutional network.

5) Swin-Tiny: Swin-Tiny was selected as a supervised hierarchical vision transformer. Unlike a standard ViT that applies global attention directly across all patches, Swin Transformer performs attention within local windows and gradually combines information through shifted windows and hierarchical stages. Transformer-based and Swin-based models have also been applied to plant-disease recognition [9]–[11].

6) DINOv2 ViT-B/14: DINOv2 ViT-B/14 was selected to represent self-supervised visual pretraining. Unlike the supervised backbones, its original representation learning does not depend on manually assigned image-class labels. Instead, general visual representations are learned from large collections of images using a self-supervised training objective.This is important for AgriHGT because the graph does not require only a strong classifier. It requires an embedding space in which visually and semantically related leaf images can be represented meaningfully. Self-supervised representations may preserve visual structure that is not restricted to the classes used during supervised pretraining. DINOv2 therefore provides a useful comparison between supervised and self-supervised feature spaces.DINOv2 therefore provides a useful comparison between supervised and self-supervised feature spaces, following the growing use of self-supervised learning in plant-disease recognition [12], [13].

7) DINOv3 ViT-B/16: DINOv3 ViT-B/16 was included as a more recent self-supervised vision transformer from the

DINO family. It was selected to examine whether a newer self-supervised representation provides stronger leaf-image features for downstream heterogeneous graph learning. The comparison between DINOv2 and DINOv3 is especially useful because both belong to the self-supervised transformer family. Differences between them therefore provide information about the effect of the quality and generation of the pretrained representation, rather than only comparing a CNN with a transformer.

8) Backbone Categories: The selected backbones can be summarized according to architecture and pretraining strategy as shown in Table III.

H. Backbone Adaptation Before Feature Extraction

The pretrained backbones were not used directly as fixed generic feature extractors. Each backbone was first adapted to the 28-class leaf-disease classification task using only the training partition. This step was performed because features learned from general image datasets may emphasize object shape, background, and general appearance rather than the small colour and texture differences required for disease classification.

A temporary 28-class classification head was attached to each backbone. Most pretrained layers were kept frozen, while only the final representation stage and classification head were fine-tuned. This allowed the high-level representation to become more disease-specific while preserving the general low- and mid-level visual features learned during pretraining.

For ResNet50, `layer4` and the classification head were fine-tuned. For DenseNet121, `denseblock4`, `norm5`, and the classifier were fine-tuned. For EfficientNet-B3, `features[7:]` and the classification head were trained. For ConvNeXt-Tiny, `features[7]` and the classification head were trained. For Swin-Tiny, the final transformer stage, terminal normalization layer, and classification head were fine-tuned. For DINOv2 and DINOv3, the final two transformer blocks, terminal normalization components, and temporary classification head were fine-tuned.

Partial fine-tuning was preferred to complete end-to-end retraining because the dataset is considerably smaller than the datasets used for backbone pretraining. Freezing most of the encoder reduces computational cost and limits unnecessary modification of useful pretrained features.

I. Feature Extraction

During backbone adaptation, validation accuracy was monitored after each epoch. Early stopping was applied to the backbones trained with a patience-based schedule, and training was stopped when validation accuracy did not improve for the predefined number of epochs. The checkpoint corresponding to the highest validation accuracy was saved during training and restored after training was completed. After the best checkpoint was restored, the temporary 28-class classification head was removed or replaced with an identity mapping. The backbone parameters were then frozen, and the adapted models were used only as feature extractors.

A deterministic feature-extraction pass was performed over all 26,866 retained images. Random training augmentation was not applied during this stage. Each image was processed using the input resolution and normalization required by the corresponding backbone.

The resulting feature matrices were:

- ResNet50: $26{,}866 \times 2{,}048$;
- DenseNet121: $26{,}866 \times 1{,}024$;
- EfficientNet-B3: $26{,}866 \times 1{,}536$;
- ConvNeXt-Tiny: $26{,}866 \times 768$;
- Swin-Tiny: $26{,}866 \times 768$;
- DINOv2 ViT-B/14: $26{,}866 \times 768$; and
- DINOv3 ViT-B/16: $26{,}866 \times 768$.

For each backbone, one feature vector was extracted from every retained leaf image. Therefore, the number of rows in the feature matrix was equal to the number of images, while the number of columns represented the embedding dimension of the corresponding backbone.

J. Feature Standardization

Feature dimensions produced by the backbones can differ substantially in scale and variance. Standardization was therefore performed before graph construction. For each backbone, the mean and standard deviation of every feature dimension were calculated using only the training partition.

For a feature value x,

$$x' = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}},$$

where μ_{train} and σ_{train} were calculated from the training embeddings. The same statistics were then applied unchanged to validation and test embeddings.

Standardization was carried out separately for each backbone. A separate heterogeneous graph was subsequently constructed for every backbone because the `similar_to` relation depends on distances in the corresponding feature space.

K. Heterogeneous Graph Construction

A heterogeneous graph was constructed from each standardized backbone feature matrix. Graph-based learning has previously been used to capture relationships among visual regions or leaf images in plant-disease recognition [23], [26], while agricultural knowledge graphs have been used to represent semantic relationships among crops, diseases, pests, symptoms, and related entities [27]–[29]. AgriHGT combines these two forms of information by representing individual leaf images together with crop and disease-related entities in a common graph. The graph structure is provided in the figure3.

Four node types were used:

- `leaf_image`: one node for each retained image;
- `crop`: one node for each of the four crop species;
- `disease`: one node for each of the 28 crop-specific healthy or disease classes; and
- `disease_category`: one node for each broader disease group.

The graph contained 26,866 `leaf_image` nodes, four `crop` nodes, 28 `disease` nodes, and seven

TABLE III
VISUAL BACKBONE CATEGORIES AND THEIR ROLE IN AGRIHGT

Backbone	Architecture	Pretraining Type	Reason for Inclusion
ResNet50	Residual CNN	Supervised	Standard CNN reference
DenseNet121	Densely connected CNN	Supervised	Feature reuse and alternative CNN representation
EfficientNet-B3	Compound-scaled CNN	Supervised	Efficient depth, width, and resolution scaling
ConvNeXt-Tiny	Transformer-inspired modern CNN	Supervised	Bridge between conventional CNNs and modern designs
Swin-Tiny	Hierarchical vision transformer	Supervised	Local-window and hierarchical transformer representation
DINOv2 ViT-B/14	Vision Transformer	Self-supervised	Strong self-supervised visual representation
DINOv3 ViT-B/16	Vision Transformer	Self-supervised	Recent self-supervised representation from the DINO family

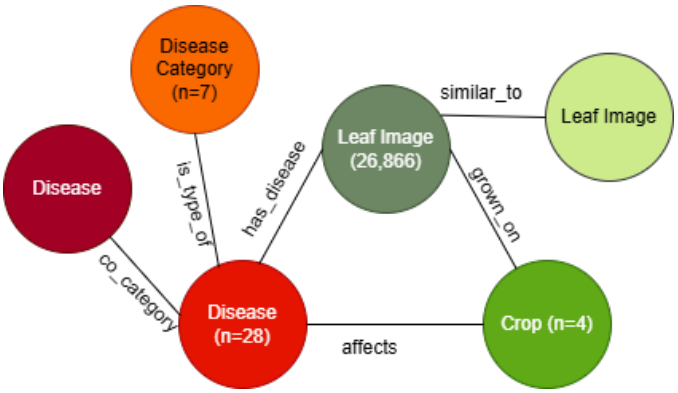


Fig. 3. Heterogeneous graph structure used in AgriHGT, showing the `leaf_image`, `crop`, `disease`, and `disease_category` node types and their typed relations.

`disease_category` nodes, giving 26,905 nodes in total.

The seven broader categories were Healthy, Viral, Bacterial, Fungal, Pest, Nutrient Deficiency, and Other. The category assignment was based on the crop-specific class name and was fixed before HGT training. Similar use of structured crop and disease relationships can be found in agricultural knowledge-graph studies [27], [28]. In AgriHGT, this categorical level was used to provide a common semantic connection among crop-specific disease classes belonging to the same broader group.

The standardized backbone vector of each image was used as the input feature of its corresponding `leaf_image` node. Crop, disease, and disease-category nodes have no pixel-derived feature vectors, so each non-image node was assigned its own learnable input embedding.

1) Graph Relations: Six forward relation types were used:

1) `grown_on`: leaf image → crop;
2) `affects`: crop → disease;
3) `is_type_of`: disease → disease category;
4) `similar_to`: leaf image → visually similar leaf image;
5) `has_disease`: selected training leaf image → disease; and
6) `co_category`: disease → another disease belonging to the same broader category.

A separate reverse relation was generated for each forward relation. A self-loop was also added to every node. The final graph therefore contained 13 relation types: six forward relations, six reverse relations, and one self relation.

2) Visual Similarity Relation: The `similar_to` edges were constructed from the standardized backbone embeddings. Connecting visually related leaf representations has also been used in graph-based plant-disease modelling, where feature-space similarity is used to define relationships among image nodes [26]. For each image, the five nearest neighbours were obtained using cosine distance,

$$d_{\cos}(\mathbf{x}_i, \mathbf{x}_j) = 1 - \frac{\mathbf{x}_i^T \mathbf{x}_j}{\|\mathbf{x}_i\|_2 \|\mathbf{x}_j\|_2}.$$

Neighbour search was restricted to images belonging to the same crop. This restriction was applied to reduce the influence of broad crop-specific differences in leaf shape and appearance on the nearest-neighbour structure. Five directed `similar_to` edges were created for each leaf image, producing 134,330 forward similarity edges.

The nearest neighbours were recomputed independently for every backbone because each backbone produced a different visual feature space.

L. Message-Passing and Supervision Partition

Connecting every supervised training image directly to its true disease node would create a label shortcut. With reverse graph relations, information could travel through the path

$$\texttt{leaf_image} \rightarrow \texttt{disease} \rightarrow \texttt{leaf_image},$$

allowing the true disease identity to return to an image whose prediction contributes directly to the classification loss.

To avoid this shortcut, the training partition was divided into two disjoint roles using a class-stratified 75/25 split. Original images and their linked augmented derivatives were kept on the same side of this partition.

The resulting subsets contained 18,881 `train_msg` images and 6,241 `train_sup` images.

The `train_msg` images retained their `has_disease` edges but were excluded from the classification loss. Their visual features could therefore provide information to disease nodes through message passing.

The `train_sup` images had no `has_disease` edge, and only these images contributed to the supervised HGT classification loss.

Validation and test images were also not assigned `has_disease` edges. Thus, `train_sup`, validation, and test leaf nodes followed the same rule with respect to direct access to their disease label.

After forward relations, reverse relations, and self-loops were included, the graph supplied to HGT contained 387,301 edges.

M. AgriHGT Model Architecture

AgriHGT contains three main components: heterogeneous message passing, gated fusion of graph and visual representations, and a leaf-disease compatibility decoder. The overall architecture of the proposed AgriHGT framework is shown in Fig. 4.

1) Node Projection and HGT Message Passing: The input representations of different node types have different forms and dimensions. All node features were therefore projected to a common 256-dimensional hidden space before message passing.

For node v with type $\tau(v)$,

$$\mathbf{h}_v^{(0)} = \tanh\left(W_{\tau(v)}\mathbf{x}_v + \mathbf{b}_{\tau(v)}\right).$$

HGT was introduced for graphs containing multiple node and relation types and uses type-dependent transformations and relation-aware attention during message passing [34]. This property is suitable for AgriHGT because a message transmitted through a leaf-similarity relation does not represent the same information as a message transmitted through a crop-disease or disease-category relation.

Four attention heads were used, the hidden dimension was 256, and HGT dropout was set to 0.2.

Three message-passing depths were evaluated:

$$n_{\text{layers}} \in \{0, 1, 2\}.$$

The zero-layer configuration performed the node-type-specific projection but skipped all HGT message-passing layers. The residual visual branch, learnable non-image embeddings, fusion mechanism, and disease decoder remained present.

The original pyHGT code still created one graph-convolution layer even when the number of HGT layers was set to 0. Therefore, the model was modified so that the zero-layer version performed no message passing at all.

2) Gated Residual Fusion: The original standardized image feature was independently projected into the 256-dimensional hidden space:

$$\mathbf{r}_i = \text{LayerNorm}\left(W_r\mathbf{x}_i + \mathbf{b}_r\right).$$

Let $\mathbf{h}_i$ denote the graph-updated representation of leaf image i. A feature-wise gate was calculated as

$$\mathbf{g}_i = \sigma\left(W_g[\mathbf{r}_i \| \mathbf{h}_i] + \mathbf{b}_g\right),$$

where $\|$ denotes concatenation.

The final leaf representation was

$$\mathbf{z}_i = \mathbf{g}_i \odot \mathbf{r}_i + (1 - \mathbf{g}_i) \odot \mathbf{h}_i.$$

The gate bias was initialized to 2.0. Since $\sigma(2) \approx 0.88$, the model initially favored the direct visual representation and learned during training how much graph information should be incorporated.

3) Leaf-Disease Compatibility Decoder: The final leaf representation was compared with the representations of all 28 disease nodes to determine the best disease match.

First, the leaf feature $\mathbf{z}_i$ and disease feature $\mathbf{d}_j$ were projected into a common 128-dimensional space:

$$\mathbf{u}_i = W_l\mathbf{z}_i + \mathbf{b}_l,$$

and

$$\mathbf{v}_j = W_d\mathbf{d}_j.$$

The transformed leaf and disease features were then combined as

$$\mathbf{u}_i + \mathbf{v}_j.$$

This combined representation was passed through a ReLU activation, dropout of 0.1, and a final linear layer to calculate a compatibility score between leaf i and disease j:

$$s_{ij} = W_o\,\text{Dropout}\left[\text{ReLU}\left(\mathbf{u}_i + \mathbf{v}_j\right)\right] + b_o.$$

A compatibility score was calculated for each of the 28 disease nodes. The disease with the highest score was selected as the final prediction:

$$\hat{y}_i = \arg\max_j s_{ij}.$$

In simple terms, the decoder measures how well the final leaf feature matches each disease representation and predicts the disease with the highest matching score.

N. Model Training and Optimization

AgriHGT was trained in full-batch mode. The entire heterogeneous graph was processed during each forward pass. Classification loss was calculated only from the 6,241 `train_sup` leaf nodes. The `train_msg`, validation, and test labels were not included in the loss.

Weighted cross-entropy was used to account for unequal class frequencies within the `train_sup` subset. For class c, the initial weight was calculated as

$$\tilde{w}_c = \frac{1}{n_c},$$

where n_c is the number of `train_sup` images belonging to class c. The weights were subsequently normalized to have a mean of one.

The training loss was

$$\mathcal{L} = -\frac{1}{|\mathcal{S}|}\sum_{i\in\mathcal{S}} w_{y_i}\log\left(\frac{\exp(s_{i,y_i})}{\sum_{j=1}^{28}\exp(s_{ij})}\right),$$

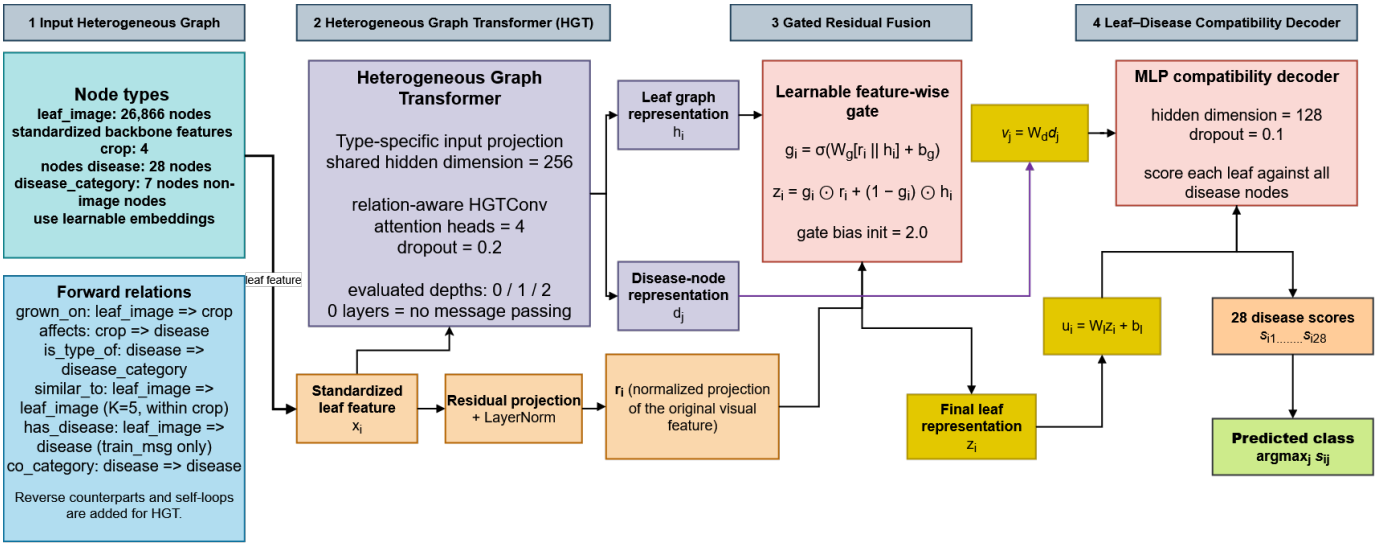


Fig. 4. Architecture of the proposed AgriHGT framework. The model combines pretrained visual feature extraction, heterogeneous graph construction, relation-aware HGT message passing, residual visual fusion, and final disease prediction.

where $\mathcal{S}$ denotes the `train_sup` set.

AdamW was used with a learning rate of 1×10^{-3} and weight decay of 1×10^{-4}. Gradient norms were clipped at 1.0. A cosine learning-rate schedule reduced the learning rate toward 1×10^{-6}.

For ResNet50, Swin-Tiny, and DenseNet121, each HGT-depth configuration was trained for a maximum of 300 epochs. The patience value was also 300, so these experiments effectively used the full training budget while retaining the checkpoint with the highest validation accuracy.

For ConvNeXt-Tiny, EfficientNet-B3, DINOv2, and DINOv3, the maximum training budget was 400 epochs with validation-accuracy early stopping using a patience of 50 epochs.

A minimum validation-accuracy improvement of 10^{-4} was required. For every backbone, zero, one, and two layer models were evaluated under the same data partition. The preferred HGT depth was determined from validation accuracy. Test accuracy and macro-F1 were reported for the individual depth configurations, but test performance was not used to select either the training checkpoint or the preferred HGT depth.

1) Transductive Training Setting: The HGT experiments used a transductive node-classification setting. Training, validation, and test leaf-image nodes were all included in the heterogeneous graph during HGT training. Consequently, the visual features and graph connections of validation and test nodes could participate in message passing.

Their true disease labels were not included in the optimization loss, and validation and test nodes were not connected to their true disease nodes through `has_disease` edges.

The reported HGT results therefore describe transductive classification. They should not be interpreted as a fully inductive evaluation in which test images are absent from the graph during training and are introduced only after the model parameters have been learned. Therefore, the reported evaluation measures how well AgriHGT classifies unlabeled leaf nodes that are already present in the graph, rather than completely unseen leaf nodes introduced after training.

O. Reference Models and Ablation Design

1) Feature-Only Logistic-Regression Reference: A multinomial logistic-regression classifier was trained separately for each backbone using the same standardized visual features but without any graph structure. Feature-based classification provides a useful reference because conventional plant-disease recognition commonly relies on visual representations without explicitly modelling relationships among individual images, crops, and disease entities [1], [2].

The model did not use crop nodes, disease nodes, disease-category nodes, graph relations, HGT layers, residual graph fusion, or the leaf–disease decoder.

All 25,122 training images were available to the logistic-regression model because no label-bearing graph edges were present and therefore no message-passing/supervision separation was required. The maximum number of optimization iterations was 2,000.

This model provides a feature-only reference for determining how much classification performance is already present in the adapted backbone embeddings. Because logistic regression uses all training labels directly whereas AgriHGT divides the training images into `train_msg` and `train_sup` roles, it should be treated as a visual-feature reference rather than a perfectly matched supervision ablation.

2) Zero-Layer AgriHGT Reference: The zero-layer AgriHGT configuration provides a closer architectural reference. It retains the node projections, learnable non-image embeddings, residual visual pathway, fusion mechanism, and disease decoder, but performs no heterogeneous message passing.

Comparison of the zero-layer configuration with the one and two layer configurations therefore provides the most direct experiment for examining the effect of HGT message propagation while retaining the remaining AgriHGT architecture.

3) Graph-Component Ablation: A component-level ablation study was conducted using the DINOv3 ViT-B/16 feature representation. The purpose of this experiment was to examine the contribution of individual graph relations and architectural components while keeping the remaining model configuration fixed.

The two-layer HGT configuration was used as the full-model reference because two HGT layers produced the highest validation accuracy in the DINOv3 depth experiment used for the ablation analysis. All ablation configurations used a hidden dimension of 256, four attention heads, HGT dropout of 0.2, the same MLP-based leaf–disease decoder, and the same class-weighted cross-entropy objective. The duplicate-aware data split, DINOv3 feature bank, `train_msg`/`train_sup` partition, validation set, and test set were kept unchanged.

Nine configurations were considered, as summarized in Table IV.

For the structural ablations A2-A5, the heterogeneous graph was rebuilt after removing only the node types or relation types specified by the corresponding configuration. All other graph-construction rules were left unchanged. In particular, the within-crop nearest-neighbour graph was retained for A3-A5, while it was omitted only for A2.

For A6, the graph structure remained unchanged, but the learnable representations assigned to crop, disease, and disease-category nodes were replaced by zero vectors. For A7, the residual visual pathway was removed and the graph-updated leaf representation was passed directly to the disease decoder. For A8, the learned gate

$$\mathbf{z}_i = \mathbf{g}_i \odot \mathbf{r}_i + (1 - \mathbf{g}_i) \odot \mathbf{h}_i$$

was replaced by

$$\mathbf{z}_i = 0.5\mathbf{r}_i + 0.5\mathbf{h}_i.$$

A0 and A2-A8 were trained for a maximum of 400 epochs using AdamW with a learning rate of 1×10^{-3} and weight decay of 1×10^{-4}. A cosine learning-rate schedule reduced the learning rate toward 1×10^{-6}. Validation accuracy was used for checkpoint selection with an early-stopping patience of 50 epochs and a minimum required improvement of 10^{-4}. Random seed 42 was used for all ablation configurations.

A1 was not trained again as a separate ablation run because the zero-layer DINOv3 configuration already provided the corresponding no-message-passing experiment under the same training settings. Its result was therefore reused in the ablation comparison.

Each ablation changed one defined component relative to the full configuration while retaining the remaining settings as consistently as possible. Because the ablation study was conducted using a single fixed random seed, the resulting differences are treated as component-level observations rather than estimates of statistical variability.

P. Evaluation Metrics

Performance was evaluated using accuracy, precision, recall, F1-score, and macro-F1. A confusion matrix was also used to examine class-level errors.

For N test images,

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{N}.$$

For class c, precision and recall were calculated as

$$P_c = \frac{TP_c}{TP_c + FP_c},$$

and

$$R_c = \frac{TP_c}{TP_c + FN_c},$$

where TP_c, FP_c, and FN_c denote true-positive, false-positive, and false-negative counts for class c.

The class-wise F1-score was

$$F1_c = \frac{2P_cR_c}{P_c + R_c}.$$

Because the class distribution was imbalanced, macro-F1 was calculated as

$$F1_{\text{macro}} = \frac{1}{28}\sum_{c=1}^{28} F1_c.$$

Macro-F1 gives the same weight to every class regardless of the number of test samples. Per-class precision, recall, F1-score, and support were also reported.

Q. Experimental Setup

All experiments were conducted in the Kaggle notebook environment using a single NVIDIA Tesla T4 GPU. PyTorch and Torchvision were used for supervised backbone training and feature extraction. DINOv2 was loaded through the official DINOv2 implementation, while DINOv3 ViT-B/16 was loaded through its `timm` implementation. The pyHGT codebase was used for heterogeneous graph representation and HGT layers, with the zero-layer behavior modified as described in Section III-M1. PyTorch Geometric supported graph-related operations.

Scikit-learn was used for nearest-neighbour search, logistic regression, and evaluation utilities. Pandas and NumPy were used for data handling, while Pillow and ImageHash were used for image processing and perceptual-hash analysis.

The duplicate-aware support-controlled split was fixed across all backbone experiments. Random seed 42 was used for the main HGT and component-ablation experiments. A fixed hold-out validation partition was used instead of k-fold cross-validation. Validation accuracy determined the retained checkpoint and the preferred HGT depth.

The main backbone experiments were single-seed experiments. The reported component-ablation experiment was also performed using a single fixed seed. Multi-seed estimates of variability were not obtained and are therefore treated as a limitation of the present study.

TABLE IV
COMPONENT ABLATION CONFIGURATIONS USED WITH DINOV3 FEATURES

ID	Configuration	Modification
A0	Full AgriHGT	Retained all graph relations and architectural components with two HGT layers.
A1	No HGT message passing	Used zero HGT layers, so no `GeneralConv` message-passing operation was performed.
A2	No leaf-similarity relation	Removed the `similar_to` and reverse-`similar_to` relations while retaining all other graph components.
A3	No crop knowledge	Removed crop nodes together with the `grown_on`, reverse-`grown_on`, `affects`, and reverse-`affects` relations.
A4	No disease taxonomy	Removed disease-category nodes together with the `is_type_of`, reverse-`is_type_of`, `co_category`, and reverse-`co_category` relations.
A5	No disease visual grounding	Removed the `has_disease` and reverse-`has_disease` relations that normally connect `train_msg` leaf nodes with their disease nodes.
A6	No learnable meta-node identity	Replaced the learnable crop, disease, and disease-category input embeddings with fixed zero vectors.
A7	No residual visual bypass	Removed the direct projected DINOv3 visual pathway so that prediction depended only on the graph-updated leaf representation.
A8	No learned fusion gate	Replaced the learned feature-wise fusion gate with a fixed equal-weight average of the projected visual and graph representations.

IV. RESULTS AND DISCUSSION

A. Experimental Results

1) Performance of Supervised-Pretrained Backbones: ResNet50, ConvNeXt-Tiny, EfficientNet-B3, Swin-Tiny, and DenseNet121 were initialized from supervised pretrained weights. Supervised CNN and transformer architectures have been widely used for plant-disease classification [3]–[5], [10]. For each backbone, the AgriHGT configuration with the highest validation accuracy was compared with a feature-only multinomial logistic-regression reference using the same standardized backbone representation. The results are shown in Table V.

The supervised-pretrained backbones did not show a consistent improvement after graph modelling. ResNet50 increased in test accuracy from 0.8844 to 0.8927, although its macro-F1 decreased from 0.7821 to 0.7698. DenseNet121 showed a small increase in test accuracy, from 0.8512 to 0.8548, with almost unchanged macro-F1. ConvNeXt-Tiny produced the same test accuracy for the two models and only a very small macro-F1 difference.

EfficientNet-B3 performed better with logistic regression in both test accuracy and macro-F1. Swin-Tiny obtained the same test accuracy in both settings, while logistic regression produced the higher macro-F1. These results indicate that heterogeneous graph message passing did not provide a uniform advantage for supervised-pretrained visual representations. Previous studies have also reported strong plant-disease classification performance using supervised CNN and transformer representations without explicit graph modelling [5], [8], [36].

2) Performance of Self-Supervised Backbones: DINOv2 ViT-B/14 and DINOv3 ViT-B/16 were evaluated separately because their pretrained representations were obtained through self-supervised learning. Self-supervised learning has also been investigated for plant-disease recognition because it can learn useful visual representations without relying entirely on labelled pretraining data [12], [13]. Their results are presented in Table VI.

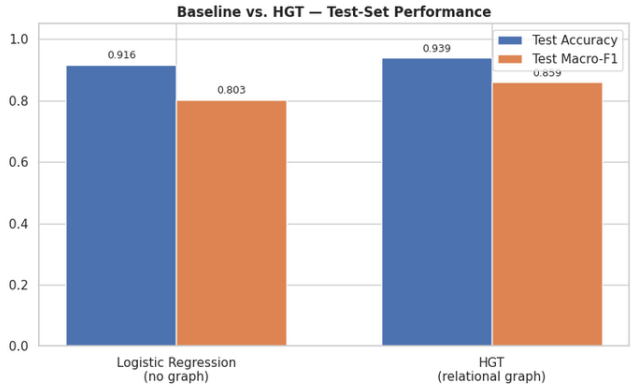


Fig. 5. Comparison of logistic regression and validation-selected AgriHGT configurations.

Both self-supervised representations showed higher test performance when used with AgriHGT. For DINOv2, test accuracy increased from 0.9175 with logistic regression to 0.9257 with one-layer AgriHGT, while macro-F1 increased from 0.8069 to 0.8374.

The largest difference was observed with DINOv3. Logistic regression achieved 0.9163 test accuracy and 0.8027 macro-F1, whereas the validation-selected two-layer AgriHGT model reached 0.9387 test accuracy and 0.8588 macro-F1. The comparison is illustrated in Fig. 5. Among all evaluated configurations, the two-layer DINOv3 AgriHGT model produced the highest test accuracy and macro-F1.

3) Effect of HGT Message-Passing Depth: Zero, one, and two layer AgriHGT configurations were evaluated for each backbone. The zero-layer configuration retained the node projections, residual visual pathway, fusion mechanism, non-

TABLE V
PERFORMANCE OF SUPERVISED-PRETRAINED VISUAL BACKBONES

Backbone	Model	HGT Layers	Val. Acc.	Test Acc.	Macro-F1
ResNet50	Logistic Regression	-	0.8962	0.8844	**0.7821**
	AgriHGT	1	**0.9118**	**0.8927**	0.7698
ConvNeXt-Tiny	Logistic Regression	-	0.8996	0.8998	0.8064
	AgriHGT	2	**0.9129**	0.8998	**0.8071**
EfficientNet-B3	Logistic Regression	-	0.8862	**0.8998**	**0.7996**
	AgriHGT	1	**0.9029**	0.8915	0.7973
Swin-Tiny	Logistic Regression	-	0.8894	0.9055	**0.8045**
	AgriHGT	2	**0.9128**	0.9055	0.7828
DenseNet121	Logistic Regression	-	0.8514	0.8512	**0.7619**
	AgriHGT	1	**0.8704**	**0.8548**	0.7616

TABLE VI
PERFORMANCE OF SELF-SUPERVISED DINO BACKBONES

Backbone	Model	HGT Layers	Val. Acc.	Test Acc.	Macro-F1
DINOv2 ViT-B/14	Logistic Regression	-	0.8951	0.9175	0.8069
	AgriHGT	1	**0.9219**	**0.9257**	**0.8374**
DINOv3 ViT-B/16	Logistic Regression	-	0.9040	0.9163	0.8027
	AgriHGT	2	**0.9263**	**0.9387**	**0.8588**

TABLE VII
EFFECT OF HGT MESSAGE-PASSING DEPTH WITH DINOV3 FEATURES

Layers	Val. Acc.	Test Acc.	Macro-F1
0	0.9107	0.9175	0.8119
1	0.9241	0.9328	0.8345
2*	**0.9263**	**0.9387**	**0.8588**

image node embeddings, and disease decoder but did not perform HGT message passing. The preferred depth for each backbone was selected according to validation accuracy.

For the supervised-pretrained representations, the validation-selected depth differed across backbones. One HGT layer was selected for ResNet50, EfficientNet-B3, and DenseNet121, whereas two layers were selected for ConvNeXt-Tiny and Swin-Tiny. The depth with the best test result was not necessarily the depth selected by validation accuracy; test performance was not used to determine model depth.

DINOv3 showed a clearer relationship between message-passing depth and performance, as reported in Table VII.

Test accuracy increased from 0.9175 with no HGT message passing to 0.9328 with one layer and 0.9387 with two layers. Macro-F1 increased from 0.8119 to 0.8345 and then to 0.8588. Validation accuracy followed the same order, resulting in selection of the two-layer model.

This experiment provides the clearest evidence in the study for a positive contribution from HGT message passing. Related graph-based plant-disease studies have also reported benefits from aggregating information among image regions or visually related leaf representations [23], [24], [26]. The result is specific to the DINOv3 representation and should not be generalized to every backbone, since several supervised-pretrained models showed mixed behaviour.

4) Training Behaviour: The training behaviour of the selected two-layer DINOv3 model is shown in Fig. 6.

Training loss decreased rapidly during the early epochs, while validation accuracy improved and reached a maximum of 0.9263. Validation loss later began to increase while training loss continued to decrease, indicating that the model was beginning to overfit the training supervision. Early stopping occurred at epoch 90, and the checkpoint corresponding to the highest validation accuracy was retained for final evaluation.

5) Class-Level Performance and Error Patterns: The confusion matrix of the selected two-layer DINOv3 model is shown in Fig. 7.

The class-wise test support and F1-scores of the selected two-layer DINOv3 AgriHGT model are reported in Table VIII.

The model classified most Bottle Gourd, Papaya, and Tomato classes reliably, with several classes obtaining F1-scores close to or equal to 1.000. The largest errors were concentrated among several Zucchini disease classes.

Zucchini Angular Leaf Spot had 36 test samples and obtained a precision of 0.684, recall of 0.361, and F1-score of 0.473. Zucchini Anthracnose obtained an F1-score of 0.381. Since Angular Leaf Spot had substantially more test samples than several minority classes, its low recall cannot be attributed only to extremely small test support.Similar lesion patterns and overlapping visual symptoms may contribute to the observed confusion, since fine-grained separation of visually similar plant diseases remains challenging [6], [10], [20].

Several other Zucchini classes had very small test support. Zucchini Healthy, Iron Chlorosis Damage, and Xanthomonas Leaf Spot contained only one or two test examples. Class-wise scores for these classes are unstable because a single correct

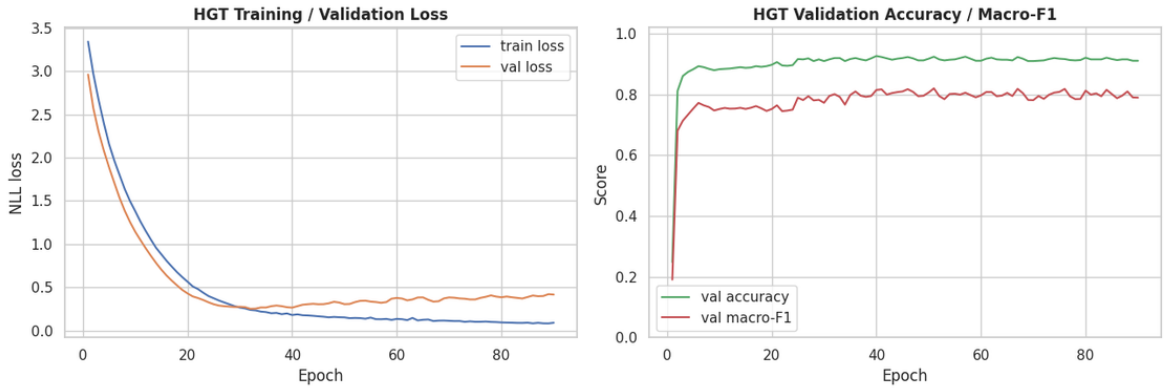


Fig. 6. Training behaviour of the selected two-layer DINOv3 AgriHGT model.

TABLE VIII
CLASS-WISE TEST SUPPORT AND F1-SCORE OF THE SELECTED TWO-LAYER DINOV3 AGRIHGT MODEL

Class	Support	F1-Score
Papaya Curled Yellow Spot	66	0.969
Papaya Carica Insect Hole	56	0.945
Bottle Gourd Mosaic Virus	52	0.971
Bottle Gourd Downy Mildew	51	0.980
Papaya Pathogen Symptoms	51	0.990
Bottle Gourd Healthy	49	1.000
Tomato Healthy	47	0.979
Bottle Gourd Alternaria Leaf Blight	46	0.978
Tomato Spot	43	0.951
Tomato Mosaic	40	0.976
Bottle Gourd Anthracnose	40	0.974
Papaya Healthy Leaf	36	0.973
Zucchini Angular Leaf Spot	36	0.473
Bottle Gourd Early Alternaria Leaf Blight	31	0.984
Papaya Bacterial Blight	31	0.984
Papaya Mosaic Virus	30	0.984
Zucchini Yellow Mosaic Virus	30	0.967
Zucchini Downy Mildew	23	0.889
Tomato White Spot	21	1.000
Tomato Downy	19	0.950
Bottle Gourd Fungal Damage Leaf	15	1.000
Papaya Yellow Necrotic Spots Holes	11	0.957
Zucchini Anthracnose	10	0.381
Zucchini Insect Damage	8	0.696
Zucchini Healthy	2	0.667
Zucchini Dry Leaf	2	0.364
Zucchini Iron Chlorosis Damage	1	0.400
Zucchini Xanthomonas Leaf Spot	1	0.667

or incorrect prediction can substantially alter precision, recall, and F1-score.

The difference between the overall test accuracy of 0.9387 and macro-F1 of 0.8588 also indicates that performance was not equally distributed across the 28 classes.

6) Qualitative Visual Analysis: Grad-CAM was used to inspect the visual regions emphasized by the DINOv3 backbone. Grad-CAM and related activation-based visualization methods have also been used in plant-disease studies to examine disease-relevant image regions [3], [8], [18].Representative examples are shown in Fig. 8. The activation maps commonly emphasized regions containing visible discoloration, spots, lesions, or damaged leaf tissue.

Selected misclassified images are shown in Fig. 9. Several errors occurred between classes with visually similar disease patterns.

These Grad-CAM maps describe the visual backbone component and should not be interpreted as explanations of HGT message passing or of the complete relational decision process. They provide only a qualitative indication of the image regions contributing to the DINOv3 visual representation.

7) DINOv3 Component Ablation: A component-level ablation study was performed using DINOv3 ViT-B/16 features to examine the contribution of the main structural and architectural components of AgriHGT. Nine configurations were considered. A0 retained the complete two-layer AgriHGT

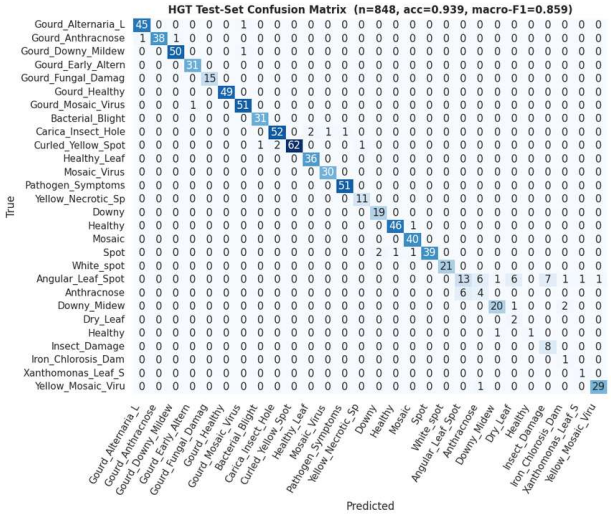


Fig. 7. Test confusion matrix of the selected two-layer DINOv3 AgriHGT model.

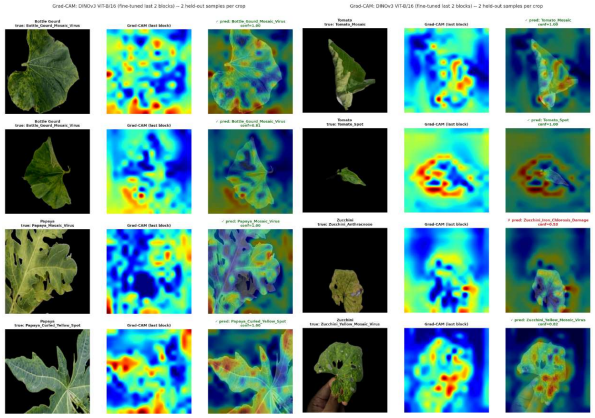


Fig. 8. Representative Grad-CAM visualizations obtained from the DINOv3 visual backbone.

model, while A1–A8 removed or replaced one component at a time.

The ablation experiments used a fixed random seed of 42. A0 and A2–A8 were trained for a maximum of 400 epochs with validation-accuracy early stopping using a patience of 50 epochs and a minimum improvement threshold of 10^{-4}. The hidden dimension was 256, four attention heads were used, HGT dropout was 0.2, and the remaining optimization settings were kept consistent with the DINOv3 HGT configuration. A1 was not retrained because the corresponding zero-layer DINOv3 run was already available and directly represented the no-message-passing configuration.

The component-ablation series was conducted as a separate DINOv3 experimental run. In this run, the full two-layer model A0 obtained 0.9328 test accuracy and 0.8219 macro-F1, while the reused zero-layer model A1 obtained 0.9281 accuracy and 0.8300 macro-F1. These values differ from the main DINOv3 depth results reported in Table VII. The ablation results should therefore be interpreted relative to A0 within Table IX, rather than compared directly with the main depth experiment.

The largest decrease in both metrics occurred in A7, where the residual visual bypass was removed. Test accuracy decreased from 0.9328 to 0.9233, while macro-F1 decreased from 0.8219 to 0.8119. This result indicates that preserving a direct pathway from the DINOv3 visual representation was useful and that the graph-updated representation alone did not retain all information needed for classification.

Replacing the learned feature-wise fusion gate with an equal-weight average in A8 also reduced both metrics. Test accuracy decreased to 0.9304 and macro-F1 to 0.8175. This suggests that adaptive fusion of the visual and graph representations was more effective than assigning the two representations fixed equal weights in this run.

Removing disease visual grounding in A5 reduced test accuracy to 0.9292 and macro-F1 to 0.8167. The `has_disease` relations connecting `train_msg` images with disease nodes therefore provided useful information under the evaluated configuration.

For A2, removing the leaf-similarity edges reduced accuracy but slightly improved macro-F1. One reason may be that visually similar leaves do not always belong to the same disease class. Similarity-based message passing can therefore help many common cases, but it may also confuse some minority classes with visually similar symptoms. Removing these edges may hurt overall accuracy while helping a few difficult classes.

For A3, removing crop knowledge showed a similar effect. Crop information gives useful general context, but it is shared by many diseases of the same crop. This broad information may help common classes while making it harder to separate visually similar diseases within the same crop. Removing crop-

TABLE IX
DINOv3 COMPONENT ABLATION OF AGRIHGT. A0 WAS OBTAINED FROM THE SEPARATE SINGLE-SEED COMPONENT-ABLATION RUN AND IS USED AS THE INTERNAL REFERENCE FOR A1-A8. ITS TEST RESULT (93.28% ACCURACY, 82.19% MACRO-F1) THEREFORE DIFFERS FROM THE VALIDATION-SELECTED TWO-LAYER DINOV3 RESULT REPORTED IN TABLE VII (93.87% ACCURACY, 85.88% MACRO-F1). ABLATION DIFFERENCES SHOULD BE INTERPRETED RELATIVE TO A0 WITHIN THIS TABLE.

ID	Configuration	Test Acc.	Macro-F1	ΔAcc.	ΔF1
A0	Full AgriHGT	0.9328	0.8219	-	-
A1	No HGT message passing	0.9281	0.8300	-0.0047	+0.0081
A2	No leaf-similarity edges	0.9257	0.8253	-0.0071	+0.0034
A3	No crop knowledge	0.9269	0.8359	-0.0059	+0.0140
A4	No disease taxonomy	**0.9340**	**0.8535**	+0.0012	+0.0316
A5	No disease visual grounding	0.9292	0.8167	-0.0035	-0.0052
A6	No learnable meta-node identity	0.9269	0.8278	-0.0059	+0.0059
A7	No residual visual bypass	0.9233	0.8119	-0.0094	-0.0100
A8	No learned fusion gate	0.9304	0.8175	-0.0024	-0.0044

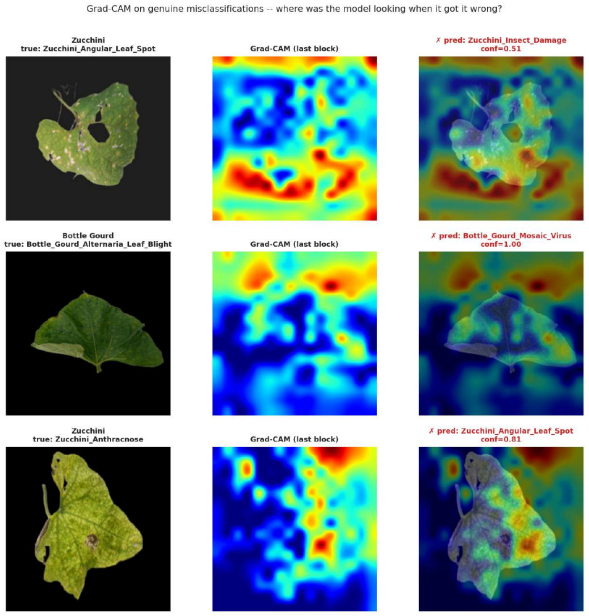


Fig. 9. Representative misclassified DINOv3 leaf images with visually similar disease symptoms.

level messages may therefore reduce total correct predictions but improve the balance across classes.

For A1, removing HGT message passing also lowered accuracy while increasing macro-F1. HGT allows information from neighbouring and semantic nodes to influence each leaf representation. This can improve overall predictions, but some propagated information may be less useful for rare or difficult classes. Without message passing, the model relies more on the leaf's own visual features, which may help some minority classes even though overall accuracy becomes slightly lower. These explanations are possible interpretations, not confirmed causes, because the ablations were run with a single seed and per-class changes were not examined for every ablation.

The strongest ablation result was obtained with A4. Removing disease-category nodes and the associated `is_type_of` and `co_category` relations increased test accuracy from 0.9328 to 0.9340 and macro-F1 from 0.8219 to 0.8535.One possible explanation is that the seven disease categories represent broad biological groups rather than visually homogeneous disease patterns. Diseases belonging to the same category can exhibit different colours, lesion shapes, textures, and affected leaf regions across different crops. Information propagated through a common disease-category node may therefore be less useful for the fine-grained visual distinctions required by the 28-class classification task. This connectivity may introduce a form of negative transfer because HGT can exchange information between diseases that share a coarse biological category but occupy substantially different regions of the visual feature space. In such cases, category-level messages may blur crop-specific disease representations and act as relational noise rather than a useful prior for fine-grained classification.

The A4 result does not establish that disease taxonomy is generally harmful. It only shows that the particular seven-category taxonomy used in this experiment did not improve performance in this single-seed ablation run. A taxonomy based more directly on symptom appearance, causal agent, affected tissue, or other disease characteristics may produce different results.

Because the ablation study used a single fixed seed, the differences in Table IX should be interpreted as component-level observations rather than statistically established effects.

B. Discussion

1) Best-Performing Configuration: The highest observed result was obtained by the two-layer DINOv3 AgriHGT model, which achieved 93.87% test accuracy and 85.88% macro-F1. The corresponding DINOv3 logistic-regression reference obtained 91.63% accuracy and 80.27% macro-F1.

Within the transductive graph setting used in this study, the DINOv3 representation therefore performed better when processed through the selected AgriHGT configuration than when classified using the feature-only logistic-regression reference. This comparison should not be interpreted as a strictly

matched message-passing ablation because logistic regression and AgriHGT use the training labels differently. Logistic regression is trained using all training images, whereas AgriHGT separates the training partition into `train_msg` and `train_sup` roles.

The zero, one, and two layer AgriHGT comparison provides a more direct examination of HGT message passing. With DINOv3 features, both validation and test performance increased from zero to one and then two HGT layers. This pattern was not observed consistently for the supervised-pretrained backbones, indicating that the usefulness of heterogeneous message passing depended on the underlying visual representation.

DINOv2 and DINOv3 were the only backbone families for which the validation-selected AgriHGT configurations improved both test accuracy and macro-F1 clearly over their logistic-regression references. Self-supervised representations may preserve feature-space structure that is useful both for leaf-node initialization and for the nearest-neighbour relations used in the graph. The present experiment, however, does not isolate the exact property of DINO representations responsible for this behaviour.

2) Interpretation of the Graph Components: The HGT-depth experiment and the component ablation provide different forms of evidence and should be interpreted separately.

The main DINOv3 depth experiment showed progressively higher test accuracy and macro-F1 from zero to two HGT layers. This provides evidence that HGT message propagation was useful for the DINOv3 representation under the main experimental setting.

The separate component-ablation series produced a more mixed pattern. Removing the residual visual pathway, learned fusion gate, or disease visual grounding reduced both test accuracy and macro-F1 relative to A0. In contrast, removing leaf-similarity edges, crop knowledge, or HGT message passing reduced accuracy while increasing macro-F1. Removal of the disease taxonomy in A4 improved both reported metrics.

The strongest AgriHGT configuration should therefore not be interpreted as evidence that every relation in the proposed graph is beneficial. Agricultural knowledge-graph studies commonly represent semantic relationships among crops, diseases, symptoms, pests, and related agricultural entities [27]–[29]. However, relations that are useful for knowledge representation are not necessarily equally useful for fine-grained visual classification. The present results suggest that the contribution of relational modelling depends on the visual representation, the message-passing process, and the particular graph relations included in the model.

3) Cross-Crop Relational Interpretation: The heterogeneous graph contains Tomato, Papaya, Zucchini, and Bottle Gourd within one shared structure. Previous studies have addressed multi-crop disease recognition using unified visual classifiers [16]–[18]. Cross-crop recognition has also been investigated using semantic attributes and domain-generalization methods [15], [19]. Crop nodes, crop-specific disease nodes, broader disease-category nodes, and leaf-image nodes are represented together.

The current experiments do not, however, establish that cross-crop semantic sharing is itself responsible for the performance improvement. The `similar_to` relation is restricted to images from the same crop, while the main explicit route for broader disease sharing is provided by the disease-category structure.

Since removal of the disease-category structure improved performance in A4, the present results provide mixed evidence for the usefulness of coarse cross-crop semantic relations. A direct comparison between a shared multi-crop graph and separately trained crop-specific graphs would be required before claiming that cross-crop relational sharing improves classification.

4) Comparison with Published Studies: Table X places AgriHGT alongside representative self-supervised, transformer-based, multi-crop, graph-based, and knowledge-fusion approaches. The numerical values should not be interpreted as a direct ranking because the studies use different datasets, class definitions, image conditions, train–test procedures, and evaluation settings.

Several published studies report higher numerical accuracy than AgriHGT, including results obtained on PlantVillage and other controlled datasets [7], [36]. Such values are not directly comparable with the Agri-Vision4 experiment because the datasets differ in class composition, field conditions, image acquisition, duplicate handling, and evaluation design.

The distinction of AgriHGT lies in its modelling structure rather than in claiming the highest accuracy across unrelated datasets. Leaf images are represented as graph nodes together with crop, disease, and disease-category entities, and relation-aware HGT layers are used to propagate information among them.

5) Error Analysis: The largest classification difficulties occurred among Zucchini disease classes. Zucchini Angular Leaf Spot obtained an F1-score of 0.473, while Zucchini Anthracnose obtained 0.381. Similar visual symptoms among disease classes may contribute to these errors, which remains a challenge in fine-grained plant-disease recognition [6], [20].

The evaluation of several Zucchini classes is also limited by very small test support. Classes containing one or two samples cannot provide stable estimates of class-wise performance. The difference between overall accuracy and macro-F1 further indicates that the strongest overall result still contains substantial variation among individual disease classes.

6) Robustness and Experimental Limitations: The main DINOv3 depth experiment and the component-ablation experiments were conducted using a single fixed random seed. The ablation configurations used a maximum training budget of 400 epochs with validation-based early stopping, but no multi-seed repetition or statistical significance testing was performed. Small differences between configurations should therefore be interpreted as observed experimental differences rather than statistically confirmed improvements.

The component-ablation series was conducted as a separate DINOv3 experimental run. Its full-model reference A0 produced different test values from the selected two-layer DINOv3 model reported in the main depth experiment. For this reason, the component-ablation results are interpreted relative to A0 within Table IX, rather than as direct numerical comparisons with the main DINOv3 result.

TABLE X
COMPARISON WITH REPRESENTATIVE PUBLISHED PLANT-DISEASE STUDIES

Study	Dataset	Method	Reported Performance	Main Difference
Wang et al. [12]	CCMT + collected dataset	SSL (MAE + CBAM)	95.35 / 99.61% Acc.	Self-supervised visual learning without graph relations
Zhao et al. [13]	PlantVillage + PlantDoc + augmented data	Contrastive SSL + domain adaptation	90.52% Acc.	Focuses on limited labels and domain shift
Thakur et al. [8]	Five public datasets	CNN + ViT	98.86% Acc. on PlantVillage	Visual hybrid without structured crop–disease relations
Subburaj et al. [16]	Five-crop datasets	Plantention	98.34% Acc., 99.02% F1	Multi-crop image classifier without graph modelling
Hasan et al. [17]	Rice, Wheat, and Bean	MobileNetV2 transfer learning	97.87% Acc.	Multi-crop classification based on visual features
Alwan and Alturfi [30]	38 plant-disease categories	EfficientNet-B8 + ViT + KGF	99.30% test Acc.	Knowledge embeddings fused with visual representations
AgriHGT	**Agri-Vision4**	**DINOv3 + HGT**	**93.87% Acc., 85.88% Macro-F1**	**Typed leaf-image, crop, disease, and disease-category graph**

The logistic-regression reference and AgriHGT also differ in their use of training labels. Logistic regression uses all 25,122 training samples as supervised examples. AgriHGT divides the training set into 18,881 `train_msg` nodes and 6,241 `train_sup` nodes so that disease-grounding edges can be used without creating a direct label shortcut. The logistic-regression comparison is consequently useful as a feature-only reference, while the zero-layer AgriHGT comparison provides the more controlled test of HGT message passing.

7) Transductive Setting and Practical Use: The current AgriHGT evaluation is transductive. Validation and test leaf nodes are present in the graph during HGT training, although their disease labels are not included in the loss and they do not receive `has_disease` edges.

The reported results therefore measure classification when the model has access to the features and graph positions of the evaluation nodes during training. They do not measure performance on completely unseen graph nodes. Cross-dataset and unseen-crop studies have shown that plant-disease models can lose performance when evaluated outside the distributions used for training [14], [15].

For deployment on new field images, an inductive procedure would be required. A new image would need to be processed by the selected backbone, standardized using training-derived statistics, inserted into the existing graph, and connected through relations that can be constructed without knowing its disease label.

Future evaluation should therefore include an inductive test in which test images are absent from the graph during model training. Additional experiments should also include multiple random seeds, greater support for minority Zucchini classes, alternative disease taxonomies, and a direct comparison between shared multi-crop and separate crop-specific graph structures.

V. CONCLUSION

AgriHGT was developed as a two-stage visual-relational framework for multi-crop leaf-disease classification using the Agri-Vision4 dataset [37]. The framework combines pretrained visual embeddings with a heterogeneous graph containing `leaf_image`, `crop`, `disease`, and `disease_category` nodes and applies relation-aware message passing through HGT [34]. Among the seven evaluated visual backbones, DINOv3 ViT-B/16 with two HGT layers achieved the highest performance, with 93.87% test accuracy and 85.88% macro-F1. The DINOv3 depth experiment showed progressive improvement from zero to two HGT layers, indicating that message passing was useful for this visual representation. However, the component ablation showed mixed effects, suggesting that the contribution of graph modelling depended on the particular relations and visual features used.

Several limitations remain. The experiments were conducted on a single regional dataset containing four crops and 28 crop-specific classes, so the findings cannot yet be generalized to other crops, diseases, geographic regions, or image-acquisition conditions. The evaluation was transductive because validation and test leaf nodes were present in the graph during training, although their disease labels were not used for optimization. Some Zucchini classes also had very small test support and showed weak class-level performance. In addition, the main experiments and the component-ablation study were conducted using a single random seed, so the observed differences were not statistically verified. The duplicate-aware splitting procedure reduced the risk of near-duplicate overlap but could not guarantee its complete absence.

Future work should evaluate AgriHGT in a fully inductive setting in which test images are introduced only after model training. Multi-seed experiments, larger datasets, and additional samples for underrepresented disease classes are also needed. Cross-dataset and unseen-crop studies have shown that performance can decrease substantially when models are evaluated outside their original training domains [14], [15], making external evaluation an important next step for AgriHGT. The disease taxonomy should also be refined, and a direct comparison between a shared multi-crop graph and separate crop-specific graphs should be conducted to determine whether cross-crop relational information provides a measurable advantage. Finally, lighter visual backbones and more efficient graph-construction methods should be investigated to improve the feasibility of deployment in resource-constrained agricultural settings.

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